

Lab 05 - Artificial Intelligence

This lab is a little different than past ones. It's going to involve practice with LLMs like chatGPT. Be sure to watch the lecture recording about how large language models work and how to use chatGPT specifically.

Upload a document that answers the following questions:

1. Make an account on <https://chat.openai.com> if you don't have one. Ask the model to generate a Python program that does *something* simple but interesting. You have freedom of choice here. Try running the program in repl.it. Include the source code and explain whether it worked as intended.

a. The prompt you used

b. The source code it produced

c. Did it work?

2. Take the following python program, which has a problem, and ask chatGPT to explain what's wrong with it and fix it. Test the code. Did it work? Include the fixed code in the document. Now ask chatGPT to re-organize the code so it has a main function that asks for the limit on the console. Does it work? Here's the broken program:

```
def sieve_of_eratosthenes(limit):
    if limit < 2:
        return []

    # Initialize a list of boolean values. All numbers are initially
    # assumed to be prime.
    is_prime = [True] * (limit + 1)
    is_prime[0:2] = [False, False] # 0 and 1 are not prime numbers

    # Iterate through the numbers and mark their multiples as
    # non-prime
    for number in range(2, limit**0.5 + 1):
```

```
    if is_prime[number]:
        for multiple in range(number*number, number, limit + 1):
            is_prime[multiple] = False

    # Collect all prime numbers
    primes = [number for number, prime in enumerate(is_prime) if
prime]

    return primes

# Example usage
limit = 200
primes = sieve_of_eratosthenes(limit)
print(primes)
```

a. The prompt you used to request the code be fixed

b. The response from chatGPT

c. Did it work?

3. What are your overall thoughts about using chatGPT for coding?